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LOOPS

Start

LOOPS

What is a Loop?
A loop is used to repeat a block of code multiple times.

Instead of writing:

```
cout << "Hello\n";
cout << "Hello\n";
cout << "Hello\n";
```

We use a loop to simplify it.

WHY USE LOOPS?

Loops help to:

- Reduce repetition
- Make code shorter
- Make programs more efficient

TYPES OF LOOPS IN C++

There are 3 main types of loops:

1. for loop → used when you know how many times to repeat
2. while loop → used when condition comes first
3. do-while loop → runs at least once

FOR LOOP SYNTAX

```
for(initialization; condition; update)
{
    // code to repeat
}
```

Parts:

- initialization → starting point
- condition → loop runs while true
- update → changes the variable

EXAMPLES

WHILE LOOP SYNTAX

```
while(condition)
{
 // code to repeat
}
```

Condition is checked before running the loop

EXAMPLES

DO–WHILE LOOP SYNTAX

```
do
{
    // code to execute
} while(condition);
```

Runs at least once, even if condition is false

EXAMPLES

DIFFERENCE

Loop Type	Condition Checked	Runs At Least Once?
for	Before	No
while	Before	No
do-while	After	Yes

INFINITE LOOP

An infinite loop happens when the condition never becomes false.

```
while(true)
{
    cout << "Hello";
}
```

Warning! This runs forever!

USING BREAK

break is used to stop a loop early.

```
for(int i = 1; i <= 10; i++)
{
    if(i == 5)
        break;

    cout << i << endl;
}
```

USING CONTINUE

continue skips the current iteration.

```
for(int i = 1; i <= 5; i++)
{
    if(i == 3)
        continue;

    cout << i << endl;
}
Output: 1 2 4 5
```

SUMMARY

- Loops repeat code
- for → known number of repetitions
- while → condition-based
- do-while → runs at least once
- Use break and continue to control loops

QUESTIONS?